AlN-capped β-$(Al_xGa_{1-x})_2O_3/Ga_2O_3$ heterostructure field-effect transistors for near-junction thermal management of next generation power devices

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While the ultrawide bandgap (E_g~4.9 eV) and high critical electric field (E_C~8 MV/cm) of β-Ga_2O_3 [1] has promising implications for power electronics, the very low bulk thermal conductivity (0.11-0.27 W/cm·K [2]) presents a formidable thermal challenge. For lateral devices, heat is typically generated within tens of nanometers of the semiconductor surface. Therefore, a pathway for efficient heat dissipation through the surface could substantially improve device-level thermal performance. In this work, we report the first experimental demonstration of top-side device-level thermal management of Ga_2O_3-based transistors by capping an $(Al_{0.21}Ga_{0.79})_2O_3/Ga_2O_3$ heterostructure field-effect transistor (HFET) with a ~400 nm thick sputter-deposited aluminum nitride (AlN) heat spreading layer. Compared to a reference HFET, we observed a ~30% reduction in device-level thermal resistance at the gate electrode.

A 125 nm thick unintentionally doped (UID) Ga_2O_3 layer followed by a 21 nm thick $(Al_{0.21}Ga_{0.79})_2O_3$ barrier layer was grown on an Fe-doped (010) Ga_2O_3 substrate via ozone-assisted molecular beam epitaxy (O_3-MBE). The $(Al_{0.21}Ga_{0.79})_2O_3$ barrier layer was delta-doped with Si ~3 nm above the heterointerface. Ohmic contacts were formed via Si ion implantation (activation anneal: 10 min, 925 °C in N_2) [3] followed by e-beam evaporation of Ti/Au (anneal: 1 min, 470 °C in N_2). The specific contact resistivity, mobility, sheet carrier concentration, and sheet resistance at room temperature were measured to be 4.6×10^{-4} Ω/cm^2, 61 cm^2/V·s, 9.55×10^{12} cm^{-2}, and 10.76 kΩ/□, respectively. For the gate dielectric, a 20 nm thick aluminum oxide (Al_2O_3) layer was deposited via atomic layer deposition (ALD). The gate electrode was formed via e-beam evaporation of Pt/Au (20/200 nm). Finally, the HFETs were capped with ~400 nm of AlN sputter-deposited at room temperature via a lift-off process. A schematic cross-section of the device structure with the AlN heat spreading layer is shown in Fig. 1. The AlN had low surface roughness and a lateral grain size of ~50 nm (Fig. 2) as determined by atomic force microscopy (AFM). Output and transfer characteristics of the AlN-capped HFET are shown in Figs. 3-5. The devices demonstrated low gate leakage (Fig. 6), and a low isolation current was confirmed (Fig. 7). Maximum attainable power density at V_{GS}=0 V was tested (Fig. 8). The majority of devices withstood a DC power density ($I_{DS}\times V_{DS}$) of ~5 W/mm before catastrophic failure; the maximum DC power density measured was ~6.8 W/mm. Thermal measurements were performed using a TMX Scientific T°Imager (532 nm, 100X objective); power dissipated was monitored using an oscilloscope, and the base temperature was maintained at 20 °C. Fig. 9 shows the temperature distribution across the length of HFETs ($L_{channel}$=15 μm) with and without the AlN heat spreading layer. The AlN-capped device shows a reduced peak temperature at the gate electrode but increased temperatures at the drain/source electrodes farther away from the channel, indicative of lateral heat spreading through the surface. Fig. 10 shows the average gate temperature rise as a function of power density. Linear fitting of this data was used to obtain device-level thermal resistance, and a ~30% reduction in thermal resistance at the gate electrode was observed for the HFET with the AlN heat spreading layer as compared to the reference device. The temperature distribution across the length of the AlN-capped HFET at power densities up to ~5 W/mm is shown in Fig. 11. The average temperature rise on the gate as a function of power density is shown in Fig. 12. Time-domain thermoreflectance (TDTR) will be used to quantify the dependence of sputtered AlN thermal conductivity on thickness and growth conditions to suggest opportunities for optimizing the heat spreading layer to further improve device-level thermal performance. OFF-state breakdown voltage testing of the AlN-capped HFETs will also be performed.

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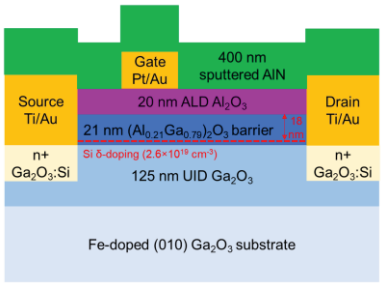


Fig. 1. Cross-section of HFET.

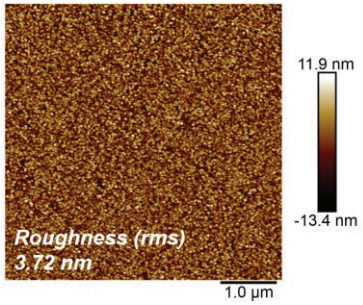


Fig. 2. AFM of sputtered AlN film.

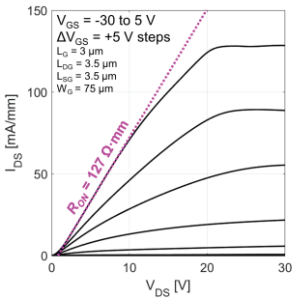


Fig. 3. DC output characteristics (I_{DS}-V_{DS}) of HFET.

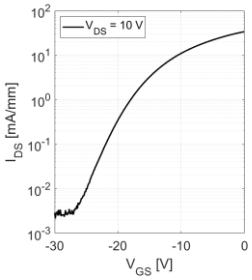


Fig. 4. DC transfer characteristics of HFET.

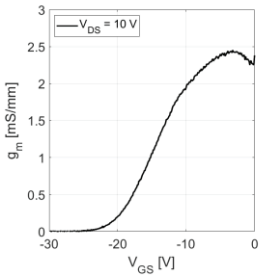


Fig. 5. DC transconductance of HFET.

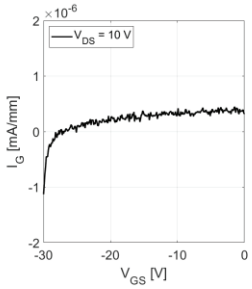


Fig. 6. Gate leakage current of HFET.

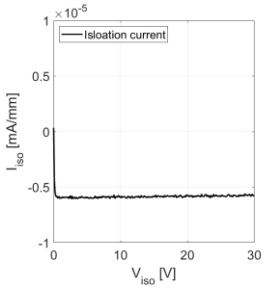


Fig. 7. Mesa isolation current of HFET.

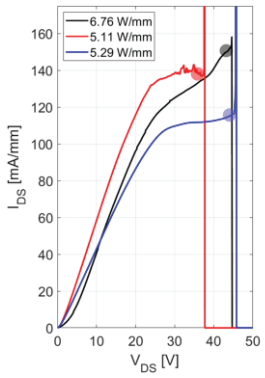


Fig. 8. Maximum power density (legend, circles) at V_{GS}=0 V.

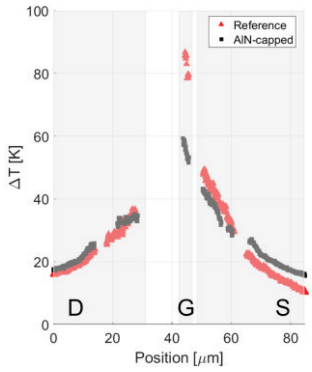


Fig. 9. Temperature rise across HFETs at 1.8 W/mm (V_{GS}=0 V). Shaded regions indicate drain (D), gate (G), and source (S) electrodes.

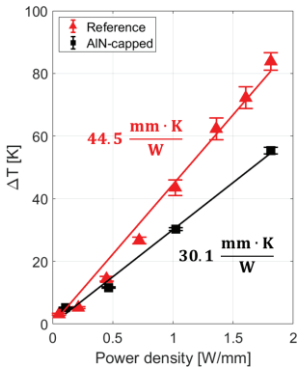


Fig. 10. Average temperature rise of gate vs. power density for HFETs. Thermal resistance is also shown.

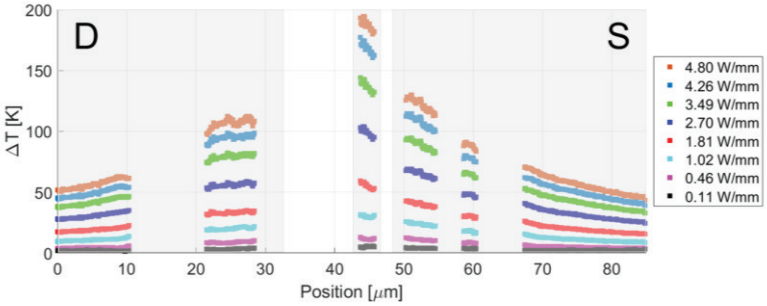


Fig. 11. Temperature rise across AlN-capped HFET as a function of power density (see legend). Shaded regions (from left to right) indicate drain (D), gate, and source (S) electrodes.

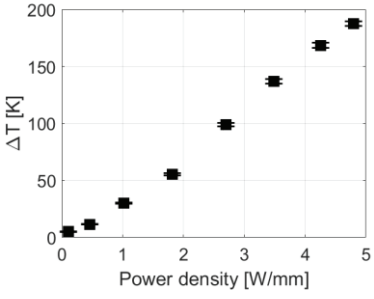


Fig. 12. Average temperature rise of gate vs. power density for AlN-capped HFET.

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